

13 Introduction to organic chemistry

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Functional groups predict characteristic reactions. Structural isomers have the same molecular formula but different connectivity; stereoisomers have the same connectivity but different spatial arrangement. Curly arrows show movement of an electron pair from its source to an electron-deficient site; distinguish electrophile, nucleophile and free radical.

Worked example: follow the reasoning

For C_4H_{10} , straight-chain butane and 2-methylpropane are chain isomers because carbon skeletons differ.

Quick check · multiple choice

A Which pair can be E/Z isomers?

[1]

A But-2-ene

B Ethene

C Propene

D 2-methylpropene

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why $CH_3CH=CHCH_3$ can have E/Z isomers but $CH_2=CHCH_2CH_3$ cannot.

[3]

Answers and explanation

Quick check: A. But-2-ene

Question 1 [3 marks]

Each alkene carbon in but-2-ene has two different groups [2]; terminal carbon of but-1-ene has two H atoms [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.